The big picture

Guiding question(s)

How can the rate of a reaction be controlled?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Different reactions proceed at different rates. The rusting of iron is a slow process whereas the burning of wood is a much faster process. The same wood when in the form of sawdust can combust explosively. A dust explosion involves the rapid combustion of a high concentration of fine dust particles suspended in the atmosphere. These particles may be flour dust, coal dust or the dust of any other flammable substance.

Dust explosions are a common hazard in industries like grain and sugar mills as well as in industries working with wood where sawdust is the main byproduct. Dust explosions are also used by special effect artists and filmmakers as they produce a spectacular display and can be contained by carefully monitoring reaction conditions.

Figure 1. The aftermath of a dust explosion.

Source "Imperial Sugar Georgia One (https://commons.wikimedia.org/wiki/File:Imperial_Sugar_Georgia_One.jpg)" by U.S. Chemical Safety and Hazard Investigation Board is in the public domain

In this subtopic we will look at the factors that affect the rate of a reaction. The study of reaction rates is called kinetics, from the Greek word *kinētikos* meaning 'motion'. Kinetics studies are of utmost importance in industry because they tell us how quickly the product will form and under what conditions. In other words, they are used to predict the economic feasibility of the reaction. What other applications of kinetics can you think of in industries like the food and beverage industry or the pharmaceutical industry?

Prior learning

Before you study this subtopic make sure that you understand the following:

- The energy changes taking place during a reaction (see subtopic R1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/energy-transfer-in-chemical-reactions-id-43469/)).
- The kinetic theory (see section S1.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/)).

Practical skills

Once you have completed this subtopic, apply your knowledge of rates of reaction by going to Practical 6a: Determining the rate of a chemical reaction (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/determining-the-rate-of-a-chemical-reaction-6a-id-46721/) and Practical 6b: Investigating factors affecting the rate of a chemical reaction (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/investigating-factors-affecting-the-rate-of-a-chemical-reaction-6b-id-46722/).